

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	CSM-B13
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: Multi-Modal AI Framework for Early Parkinson's Disease Screening Using Voice, Handwriting, Gait, Tremor, and Facial Analysis

Abstract

Parkinson's Disease (PD) is a progressive neurological disorder that affects movement, speech, and motor coordination. Early diagnosis is crucial for effective treatment and improved quality of life. However, many patients, especially those in rural and underserved areas, lack timely access to neurologists and specialized diagnostic facilities. Consequently, Parkinson's Disease is often identified only after significant motor symptoms develop. This project addresses the need for an accessible, non-invasive, and AI-assisted screening system for early Parkinson's Disease detection.

The proposed system is intended for individuals at risk of Parkinson's Disease, elderly patients, neurologists, hospitals, and telemedicine providers. Existing diagnostic methods, including neurological examinations and imaging techniques, require specialized expertise and are often expensive and inaccessible. Recent research has shown that changes in voice patterns can serve as early indicators of Parkinson's Disease. However, most existing AI-based systems rely solely on voice analysis, limiting their diagnostic reliability.

This research proposes a Smart Parkinson's Disease Detection System that analyzes voice recordings by extracting acoustic features such as MFCC, pitch, jitter, shimmer, and Harmonics-to-Noise Ratio (HNR) to estimate Parkinson's risk using AI. As a future enhancement, the system is designed to support a Multi-Modal AI Diagnosis framework by incorporating handwriting analysis, gait analysis, hand tremor detection, and facial expression analysis. Predictions from these modules can be combined using an ensemble AI model to generate a comprehensive risk assessment. The application also provides an overall risk score, explainable AI insights, prediction history, and general health recommendations while clearly stating that it is an AI-assisted screening tool and not a substitute for professional medical diagnosis.

The project is limited to non-invasive screening and risk prediction and does not provide a confirmed medical diagnosis or prescribe medication. The expected outcome is an accurate, user-friendly screening system that supports early identification of Parkinson's Disease and assists healthcare professionals in timely clinical evaluation.

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References (Category specific)

Base Paper

1. Tougui, M. Zakroum, O. Karrakchou, et al., "Transformer-Based Transfer Learning on Self-Reported Voice Recordings for Parkinson's Disease Diagnosis," *Scientific Reports*, vol. 14, Article 30131, 2024. DOI: 10.1038/s41598-024-81824-x. (Base Paper)

Supporting Papers

2. Osmar Pinto Neto, "Harnessing Voice Analysis and Machine Learning for Early Diagnosis of Parkinson's Disease: A Comparative Study Across Three Datasets," *Journal of Voice*, 2024. DOI: 10.1016/j.jvoice.2024.04.020.
3. N. D. Pah, V. Indrawati, A. Prayitno, and D. Kumar, "The Effectiveness of Acoustic Voice Quality Index to Identify People With Parkinson's Disease," *IEEE Access*, vol. 12, 2024. DOI: 10.1109/ACCESS.2024.3498836.
4. K. D. Raj, G. J. Lal, E. A. Gopalakrishnan, V. Sowmya, and J. R. Orozco-Arroyave, "A Visibility Graph Approach for Multi-Stage Classification of Parkinson's Disease Using Multimodal Data," *IEEE Access*, vol. 12, pp. 87077–87096, 2024. DOI: 10.1109/ACCESS.2024.3416444.
5. Matthew Shen, Pouria Mortezaagha, and Arya Rahgozar, "Explainable Artificial Intelligence to Diagnose Early Parkinson's Disease via Voice Analysis," *medRxiv*, 2024. DOI: 10.1101/2024.09.29.24314580.

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